

TOCOLYTIC AND OXYTOCIC DRUGS

BIETE LUNDAU LUKE

DipPharm,BPharm,MclinPharm

OXYTOCIC DRUGS

- ▶ These are drugs which have the power to excite contractions of the uterine muscle
- ▶ There are three major groups
 - Oxytocin
 - Ergot derivatives
 - Prostaglandin

Mechanism of action

- ▶ Oxytocin receptor is a G – protein coupled receptors which is connected to phospholipase C cascade
- ▶ Activation of oxytocin receptors on the myometrium triggers a downstream cascade that leads to increased intracellular calcium in uterine myofibrils which strengthens and increases the frequency of uterine contractions
- ▶ In humans, most hormones are regulated by negative feedback; however, oxytocin is one of the few that is regulated by positive feedback where more contractions promotes more oxytocin production
- ▶ Later oxytocin causes local synthesis of prostaglandin (PG-F2 α)
- ▶ Finally this leads to physiological uterine fundal contraction while enhancing cervical relaxation on the other side and thus resulting in a labor

Indications

- ▶ Induction of labor
- ▶ Labor augmentation
- ▶ Control of post partum hemorrhage

Ergot Derivatives

Examples – Ergometrine, Methylergometrine

- ▶ Acts through serotonin receptors (5-HT₂) and α adrenergic receptor
- ▶ During pregnancy, the uterine sensitivity to ergot increases dramatically
- ▶ Causes rhythmic contraction of uterus at low doses

Indications

- ▶ Prophylactic use : prevention of post partum hemorrhage
- ▶ Control of bleeding after delivery

NB:

Ergot derivatives must never be used in pregnancy, induction or augmentation of labor and during first and second stage labor prior to the delivery of the anterior shoulder

This because they have a higher potent uterotonic activity, which due to their vasoconstrictive effect, causes the increased risk of hypertension and post partum bleeding

Other side effects with ergot alkaloids are pain after birth, nausea and vomiting

Misoprostol

- ▶ This is a synthetic prostaglandin E1 analogue
- ▶ Misoprostol is administered to mimic the effect of endogenous prostaglandin

Mechanism of action of Misoprostol

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- ▶ Like the endogenous prostaglandin, misoprostol binds to smooth muscle cells in the uterine lining
- ▶ Changes in permeability and alteration in membrane bound calcium sensitizes the myometrium to oxytocin
- ▶ Further vasoconstriction in the myometrium will trigger the secretion of prostaglandin which enhances contraction
- ▶ Misoprostol increase the strength and frequency of contractions as well as degrade collagen and reduce cervical tone

Indications of Misoprostol

- ▶ Induction of labor
- ▶ Induction of medical abortion
- ▶ Management of atonic post partum haemorrhage

TOCOLYTIC DRUGS

- ▶ Drugs that prevent preterm labor and immature birth by suppressing uterine contractions (TOCOLYSIS)
- ▶ This delay in labor and delivery helps to improve perinatal outcome and hence reduce perinatal or maternal morbidity and mortality

Agents used to delay premature uterine activity include ;

- ▶ Magnesium sulphate
- ▶ NSAID - indomethacin
- ▶ Calcium channel inhibitors/blockers – nifedipine, nicardipine
- ▶ Oxytocin receptor antagonist - atosiban
- ▶ Adrenergic beta-receptor agonists – salbutamol, terbutaline, retodrine
- ▶ Nitric oxide donors - nitroglycerine

Indication of tocolytics

- ▶ Management of pre term (Premature) labour which is labour before the 37th week of gestation
- ▶ The tocolytic effect relaxes the uterus and maintains pregnancy for 24 – 48hours the period enough to have corticosteroid administered to prevent neonatal respiratory distress

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